

William S. Cleveland

Visualizing Data

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To Lisa, Bob, and Marylyn

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Preface

Visualization is critical to data analysis. It provides a front line of attack, revealing intricate structure in data that cannot be absorbed in any other way. We discover unimagined effects, and we challenge imagined ones.

Tools

Tools matter. There are exceptionally powerful visualization tools, and there are others, some well known, that rarely outperform the best ones. The data analyst needs to be hard-boiled in evaluating the efficacy of a visualization tool. It is easy to be dazzled by a display of data, especially if it is rendered with color or depth. Our tendency is to be misled into thinking we are absorbing relevant information when we see a lot. But the success of a visualization tool should be based solely on the amount we learn about the phenomenon under study. Some tools in the book are new and some are old, but all have a proven record of success in the analysis of common types of statistical data that arise in science and technology.

Graphing and Fitting

There are two components to visualizing the structure of statistical data — graphing and fitting. Graphs are needed, of course, because visualization implies a process in which information is encoded on visual displays. Fitting mathematical functions to data is needed too. Just graphing raw data, without fitting them and without graphing the fits and residuals, often leaves important aspects of data undiscovered. The visualization tools in this book consist of methods for graphing and methods for fitting.

Applications

The book is organized around applications of the visualization tools to data sets from scientific studies. This shows the role each tool plays in data analysis, and the class of problems it solves. It also demonstrates the power of visualization; for many of the data sets, the tools reveal that effects were missed in the original analyses or incorrect assumptions were made about the behavior of the data. And the applications convey the excitement of discovery that visualization brings to data analysis.

The Legacy of the Past

The visualization of statistical data has always existed in one form or another in science and technology. For example, diagrams are the first methods presented in R. A. Fisher's *Statistical Methods for Research Workers*, the 1925 book that brought statistics to many in the scientific and technical community [38]. But with the appearance of John Tukey's pioneering 1977 book, *Exploratory Data Analysis*, visualization became far more concrete and effective [76]. Since 1977, changes in computer systems have changed how we carry out visualization, but not its goals.

Display Methods

When a graph is made, quantitative and categorical information is encoded by a display method. Then the information is visually decoded. This visual perception is a vital link. No matter how clever the choice of the information, and no matter how technologically impressive the encoding, a visualization fails if the decoding fails. Some display methods lead to efficient, accurate decoding, and others lead to inefficient, inaccurate decoding. It is only through scientific study of visual perception that informed judgments can be made about display methods. Display methods are the main topic of *The Elements of Graphing Data* [20]. The visualization methods described here make heavy use of the results of *Elements* and other work in graphical perception.

Prerequisites

The reader should be familiar with basic statistics and the least-squares method of fitting equations to data. For example, an introductory course in statistics that included the fundamentals of regression analysis would be sufficient.

How to Read the Book

For most purposes, the chapters need to be read in order. Material in later chapters uses tools and ideas introduced in earlier chapters. There are two exceptions to this general rule. Chapter 6, which is about multiway data, does not use material beyond Section 4.6 in Chapter 4. Also, sections of the book labeled “For the Record” contain details that are not necessary for understanding and using the visualization tools. The details are meant for those who want to experiment with alterations of the methods, or want to implement the methods, or simply like to take in all of the detail.